

MEASUREMENT OF THE REGENER–PFOTZER MAXIMUM USING DIFFERENT TYPES OF IONISING RADIATION DETECTORS AND A NEW TELEMETRY SYSTEM TF-ATMON

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Stratospheric balloons are a useful tool for the investigation of cosmic radiation at high altitudes and the tests of new detectors of cosmic radiation. Due to necessary data processing, the balloon gondola needs to carry, together with radiation detectors, additional supplementary sensors measuring humidity, temperature, location and orientation, altitude, atmospheric pressure, acceleration, etc. A newly developed universal system TF-ATMON, based on using already existing tools of the PX4 open-source project, enables apart from data recording and monitoring, also the possibility to trace the balloon gondola after the flight. The application was demonstrated on stratospheric balloon flight FIK-6. This flight was unique because three different types of radiation detectors were used at one flight. It enabled us to compare the altitude of the Regener–Pfofzer maximum measured with different types of sensors sensitive to a different type of secondary cosmic radiation generated in the atmosphere.

INTRODUCTION

Particles of primary cosmic radiation with sufficient energy interact in the upper part of the atmosphere and generate showers of secondary cosmic radiation. With increasing depth of the atmosphere, the intensity of primary radiation decreases, whereas the secondary component increases. At an altitude of about 20 km the intensity of secondary cosmic radiation reaches its maximum, called the Regener–Pfofzer (R-P) maximum^(1, 10–11). The maximum varies with geomagnetic vertical cutoff rigidity and with solar cycle and it is generally located at 15–27 km above sea level⁽³⁾. In the past, several experiments were done with the aim to measure vertical profile of ionisation in the atmosphere at various locations in the world, mainly using radiosondes consisting of Geiger tubes^(3, 6–9), or to characterise instruments' response used for space-based missions^(2, 4, 5). Stratospheric balloons are a useful tool for the investigation of cosmic radiation at high altitudes (around and above the R-P maximum region) and for the tests of new detectors of cosmic radiation. However, the radiation measuring instruments need to be supplemented by other sensors measuring temperature, pressure, humidity, altitude, acceleration, etc. All these sensors should be continuously monitored during the launch and the flight of the balloon to verify their proper function and their values have to be recorded for further processing of all obtained data.

The universal system TF-ATMON, based on the use of existing tools of the open-source project PX4 and supplemented with an telemetry transmitter, enables it to monitor and record data during the flight and to trace the balloon. Its application is demonstrated on stratospheric balloon flight FIK-6. The experiment setup is shown in Figure 2. Various radiation detectors were used to measure the vertical profile of cosmic radiation in the atmosphere—Geiger–Müller (G-M) tubes, Si-diode-based detector SPACEDOS and scintillation detectors AIRDOS-C with inorganic scintillation crystals.

MATERIALS AND METHODS

Since 2015 we launched several stratospheric balloons, the overview of flights is summarised in Table 1. The construction of balloon gondolas was designed for specific radiation detectors during previous flights. Balloon avionics was therefore built around the chosen detectors. This concept led to a situation when every new flight meant a significant amount of work despite the fact that many components were recycled every year and used for the next one. The reason was the need to adapt the avionics to the innovative version of detectors.

Due to a relatively high value of payload it was necessary to ensure the return of the gondola

Table 1. Table summarising success of used technologies, with colours representing a degree of reliability.

	Failure	Partial failure	Correct function			
Flight	FIK-1	FIK-2	FIK-3	FIK-4	FIK-5	FIK-6
Location (year)	CZ (2015)	CZ (2017)	CZ (2018)	SE (2019)	CZ (2019)	CZ (2020)
Payload	Candy detector, Web camera	Candy detector, Web camera	AIRDOS, 360 deg camera	AIRDOS-C CRY19, SPACEDOS, G-M, Socrat-R	AIRDOS-C CRY19, SPACEDOS, G-M, 360 deg camera forest	AIRDOS-C NaI(Tl), SPACEDOS, G-M, Ionmeter, 360 deg camera
Landing site	vineyard Austria	rapeseed field	Poland	swamp (Finland)		railway corridor
Power source	Li-ion 18650 accu	Li-ion 18650 and li-pol accu	Lithium primary cells and li-pol accu	Lithium primary cells	Lithium primary cells	Li-ion 18650 accu
Telemetry system	GSM	GSM, 868 MHz Proprietary Modem	SigFox, 868 MHz Proprietary Modem	Outsourced	LoRa, SigFox, SiK 433 MHz	2x LoRa, SiK 433 MHz
Rescue beacon	433 MHz CW	433 MHz CW	433 MHz CW	Outsourced	433 MHz CW	433 MHz CW
Flight control computer	Odroid-U2	Odroid-U2	Not used	Not used/outsourced	PX4, FMU v5	PX4, FMU v5

every flight. The main construction criteria were as follows:

- Reliable transmission of information about the geographical location of the gondola
- Good resistance to impact
- Ensuring the function in temperatures far below zero

Despite the generally successful nature of all flights and the fact that the gondolas were always found, different technologies were tried to eliminate the partial shortcomings that emerged during balloon flights. During the last experimental flights these also included the way the data were recorded.

For this reason, the originally used telemetry system was made significantly more robust and an Internet of Things (IoT) LoRa (low-power long range wide-area network modulation technique) transmitter was added to the system, making it possible to transmit the data necessary for tracing the gondola to the TheThingsNetwork. This way, a high reliability of finding the gondola and recording the data was ensured. In case of active detectors, it was also necessary that the flight trajectory was recorded synchronously with the supplementary quantities.

An overall overview of the success of the used technologies is summarised in the Table 1.

Apart from technologies used in gondolas a number of supplementary tools have undergone intensive development. For example, in order to find the balloon it was necessary to have an accurate real-time map of its position together with a prediction of the subsequent flight and the site of impact. In

the case of the last two flights, FIK-5 and FIK-6, the habhub an online tool that tracks payloads in real time⁽¹⁶⁾ was used to track and predict the movement of the stratospheric balloon. As can be seen from the Table 1, during the last flights the avionics was implemented using an Unmanned Aerial Vehicle (UAV) technology. It used the Pixhawk autopilot with PX4 firmware. Telemetry was implemented by a very reliable combination of LoRa modem and SiK modem. Power was provided by Li-ion 18650 batteries that also have a reliable flight history.

Universal avionics

Based on our experiences we have used the universal concept of avionics called TF-ATMON that makes it possible to connect different types of payloads and carry out various atmospheric measurements. Furthermore, it provides the detectors with basic services such as power supply, time, position and orientation information. Basic quantities that affect many types of measurements are also recorded, including temperature, pressure, humidity, magnetic field and acceleration. Flight computer provides a possibility to record data from experiments in a common log file. Therefore, it is extremely useful for testing high-altitude cosmic radiation detectors and dosimeters. It also reduces the number of modifications in experiments (dosimeters) required for balloon flight.

The schematic diagram of the new avionics is summarised in the Figure 1.

Transition to the concept where the balloon-specific parts of avionics are completely separate from

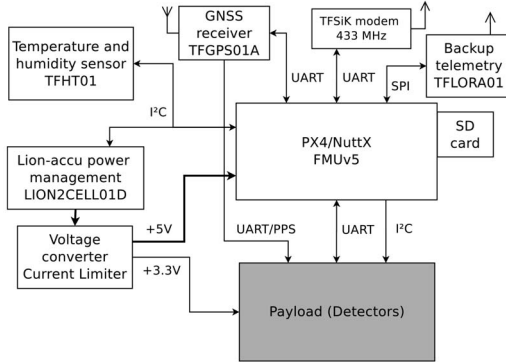


Figure 1. The schematic diagram of the new avionics.

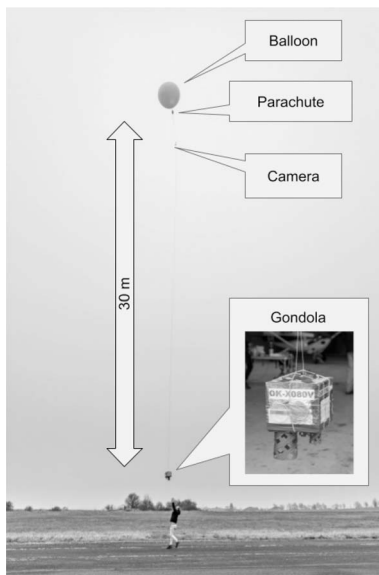


Figure 2. FIK-6 experiment setup using the Hwoyee Weather Balloon 1600.

the system of detectors simplified the realisation of next balloon flights. It reduced the complexity of connecting different types of detectors and at the same time it improved the integrity of supplementary data measurements. Overall, the new features can be summarised as follows:

- Easy implementation of different payloads
- Redundant telemetry links
- Gondola orientation and spatial position tracking and logging
- Reliable IMU sensor processing and calibration
- Possible use of relatively high-power consumption payloads
- Pre-flight continuous charging as an option

- Power monitoring and maximum uptime calculation relevant to actual temperature
- Real-time pre-flight payload diagnostic

The documentation of used blocks can be found in the following sources TFGPS01⁽¹⁷⁾, TFSiK01⁽¹⁸⁾, TFHT01⁽¹⁹⁾, TFLORA01⁽²⁰⁾, LION2CELL01D⁽²¹⁾.

Payload

In the case of FIK-5 and FIK-6 flights that served as the test flights for our novel approach of using TF-ATMON technology, the payload was not fully optimised for this use yet. All the detectors thus had their own SD cards for data recording and some even had their own power supply; therefore, the payload weight was higher than theoretically required and some lift was wasted. This situation originated in conservative flight plan, which required successful log and function of payload even in failure of the new TF-ATMON system. Results from flight FIK-6 will be presented.

Detectors

The payload of FIK-6 flight contained TF-ATMON and three different types of ionising radiation detectors: SPACEDOS with silicon PIN diode sensor (namely SPACEDOS02A), AIRDOS-C with scintillation crystal and silicon photomultiplier and a G-M tube. The total payload mass was 2 kg.

SPACEDOS is a lightweight dosimeter intended for space applications and measurements on board spacecraft. The detector has been described in⁽¹²⁾. The detector is a silicon PIN diode with a volume $300\mu\text{m} \times 10\text{ mm} \times 20\text{ mm} = 0.06\text{ cm}^3$. The energy range of this detector is from 0.2 to 9 MeV.

AIRDOS-C is a scintillation detector with a small crystal. The detector has been described in⁽¹³⁾. This detector encloses small cylindrical NaI(Tl) crystal with a volume $\varnothing 10\text{ mm} \times 20\text{ mm} = 1.6\text{ cm}^3$ and SiPM (Silicon Photomultiplier) with a sensitive surface $6\text{ mm} \times 6\text{ mm}$. The energy range of this detector is from 0.2 to 18 MeV.

The G-M tube STS-5 was used in the G-M counter. The cylindrical volume of the tube is $\varnothing 10\text{ mm} \times 76\text{ mm} = 6\text{ cm}^3$. This detector is capable of registering the flux only.

All detectors, together with other sensors and the TF-ATMON system, were put inside a polystyrene box.

RESULTS AND DISCUSSION

The flight FIK-6 took place on 18 December 2020 and lasted 1 hour and 40 minutes. The balloon was launched from Příbram airport that is located around

latitude 50° N. The balloon flight path continued in eastern direction for about 80 km.

The system TF-ATMON recorded temperature, air pressure, humidity and radiation characteristics as histograms of deposited energy of radiation events from all three radiation sensors in the gondola (Figure 3). The barometric altitude was calculated using the International Standard Atmosphere model 1976⁽¹⁴⁾.

The Figure 4 demonstrates the importance of telemetry data during measurements processing. The graphs show an increase in the response of the silicon ionising radiation detector at the times of take off, burst and landing, when there was a rapid increase in mechanical stress. The effect is probably caused by the microphone effect of the silicon detector. Telemetry data provided by the TF-ATMON system must be therefore taken into account during data evaluation. The effect of these data on measured quantities will be a subject of further research.

The graphs show that the maximum reached altitude was approximately 33 km above sea level. During the flight the balloon passed the R-P maximum twice. The rescue team followed the balloon along the whole flight trajectory. The precision of tracking allowed some participants of the rescue team to actually see the gondola touchdown visually. Therefore, the gondola was successfully rescued within few minutes after touchdown Figure 5.

By processing the data a graph of detected altitude Figure 6 was obtained. It shows that the measured altitude of the R-P maximum was for all detector types around 19 km above sea level.

The log-norm distribution fit was used to determine the R-P maximum. The fit was applied to measured data and normalised to the maximum number of particles for a given detector:

$$A \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right). \quad (1)$$

The fit parameters A , μ , σ and the calculated R-P maximum positions are given in Figure 6. The range of barometric altitudes 7000–29 000 m was chosen for the fit. The lower limit was chosen so that the course of the function was not affected by terrestrial radiation and radon progenies in the atmosphere. The upper limit was chosen based on a simulation because, according to the simulation, at higher altitudes the flux of ionising radiation does not decrease evenly.

Note that the absolute number of measured particles corresponds to the active volume, material and sensitivity of the used detector. At the same time, every detector measured a different position of R-P maximum. The lowest altitude (18 981 m) was measured by the semiconductor detector with the

density (Si, $\rho = 2.33 \text{ g/cm}^3$), followed by (18 982 m) the scintillation detector (NaI(Tl), $\rho = 3.67 \text{ g/cm}^3$) and finally, the highest altitude (19 575 m) of the maximum was measured by G-M tube (thin metallic tube filled with low-pressure gas, the exact material composition of this tube is not known).

Altitudes of maxima of all three data sets were found using the log-normal fit (Figure 6) described by mean value (μ_1, μ_2, μ_3), variance ($\sigma_1, \sigma_2, \sigma_3$) and extent (n_1, n_2, n_3) where indexes 1, 2 and 3 refer to G-M counter, NaI(Tl) scintillator and silicon PIN diode, respectively. We can see that the estimated altitudes of maxima differ. Statistical significance of this difference was evaluated using the two-selective testing of three hypotheses ($\mu_1 = \mu_2, \mu_1 = \mu_3$ and $\mu_2 = \mu_3$) about the equality of the mean values. We applied a two-selective T -test, see Table 2, which is correct also in case of a log-normal distribution according to⁽¹⁵⁾.

According to the results of the statistical analysis in Table 2, statistical criterion empirical values defined as equation 2.

$$t_{exp} = \frac{\mu_1 - \mu_2}{\sqrt{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}} \sqrt{\frac{n_1 n_2 (n_1 + n_2 - 2)}{n_1 + n_2}} \quad (2)$$

were not the element of the critical domains W defined as 3, using degrees of freedom $\nu = n_1 + n_2 - 2$

$$W = (-\infty; -t_\nu(\alpha/2)) \cup (t_\nu(\alpha/2); \infty). \quad (3)$$

The H_0 hypothesis was therefore accepted in all tested cases, except for the case of R-P maxima measured with G-M counter and silicon PIN diode, where the difference is statistically significant for significance level $\alpha = 30\%$.

As the measured data show in Figure 3, there is a very noticeable difference between a number of data points measured during the flight upwards and during the descent. This difference is mainly caused by the different value of vertical speed. In the following balloon flights we are planning to overcome this problem by a controlled descent, during which its rate can be decreased in some phases of flight so that it can be more comparable with the speed of the ascent.

At the same time, it can be seen in Figure 4, there are considerable vibrations, rapid changes in acceleration and gondola rotation during the balloon descent. The increase in humidity is also observable, which can even freeze on the instruments during some parts of the flight. All of these may affect the measurement of atmospheric quantities and for some types of instruments, they have to be compensated. However, it is

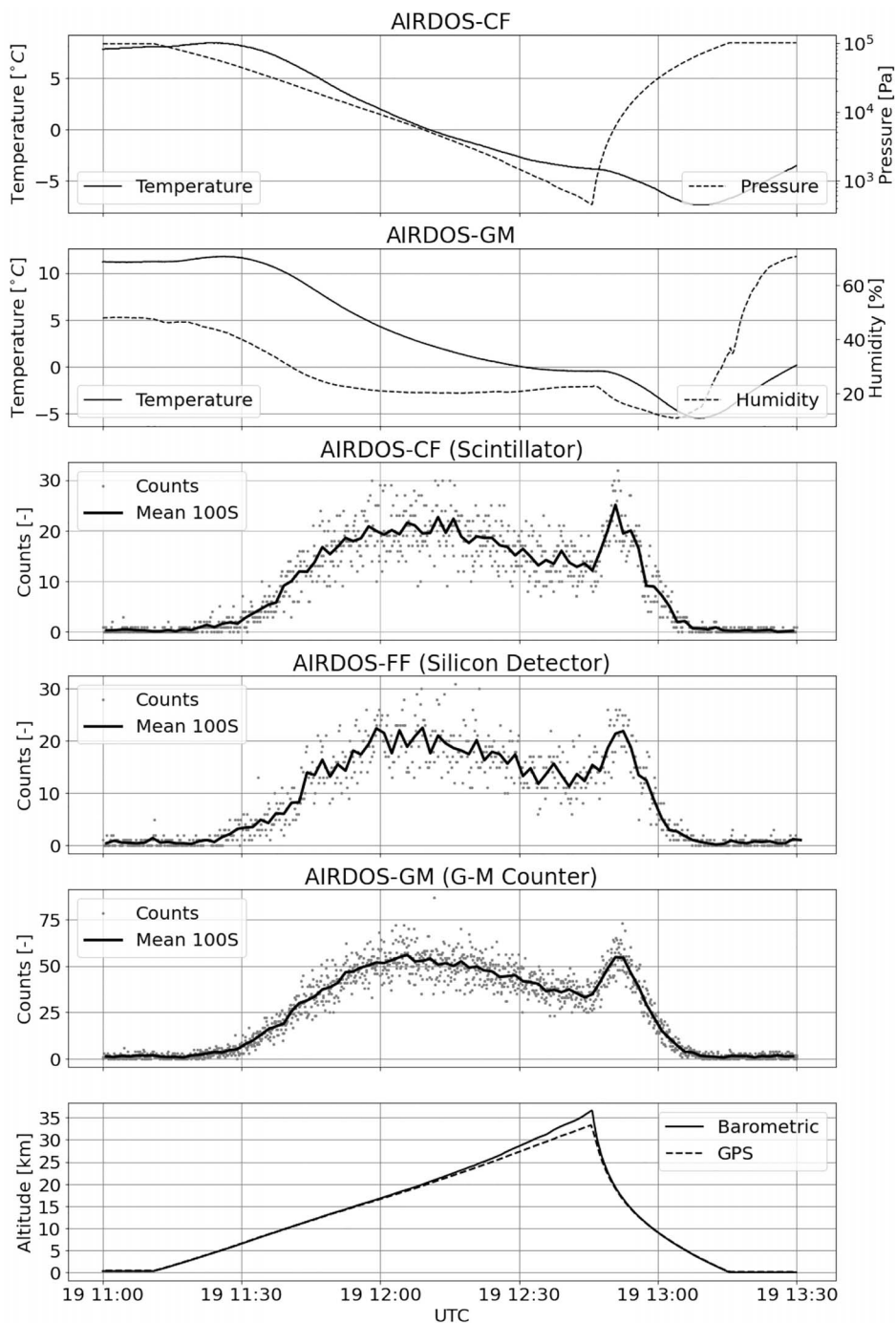


Figure 3. Raw data measured during the whole flight. From top to bottom: temperature near scintillation crystal, air pressure inside the box of crystal, temperature inside the gondola, relative humidity inside the gondola, counts of radiation events per 10 seconds counted by scintillator, silicon detector and G-M counter, barometric altitude and altitude from Global Navigation Satellite System (GNSS).

TF-ATMON IONISING RADIATION MEASUREMENT

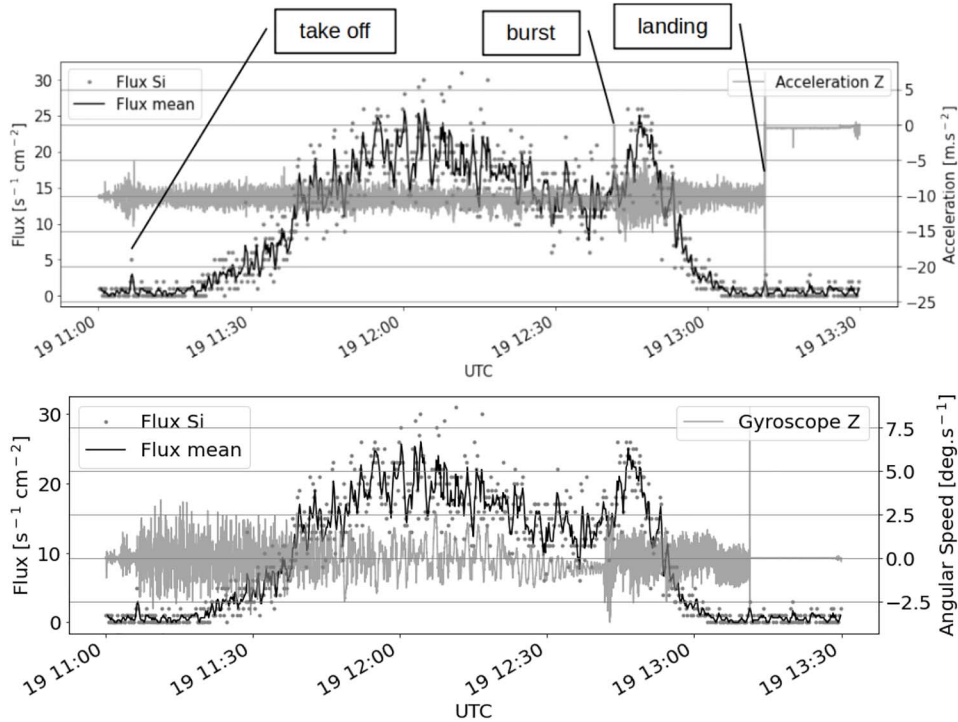


Figure 4. Acceleration and angular speed in axis perpendicular to the ground combined with ionising radiation flux measured by the silicon PIN diode detector.



Figure 5. Chasing cars with telemetry receivers at the landing site (screenshot from mobile Habhub tracking system).

not expected to be important at the current level of data processing.

It should be noted that the radiation measurements presented in Figure 6 are raw data. The correction on external effects like temperature should be applied for a correct interpretation of the measured R-P maximum altitude. Time course of the temperature is also available thanks to the records of the TF-ATMON system. The temperature dependencies of the detectors will be the subject of further studies.

Further plans

As can be seen from the Table 1, the last unsolved problem with the balloon flights is the landing site. Therefore, in the future flights we are planning to use the autopilot for a controlled descent as well. The descent will be carried out using an unmanned autogyro carrying a payload. Thus, it would be possible to choose the landing site and reduce the possible risk of creating dangerous situations and at the same time it would be possible to control the descent rate.

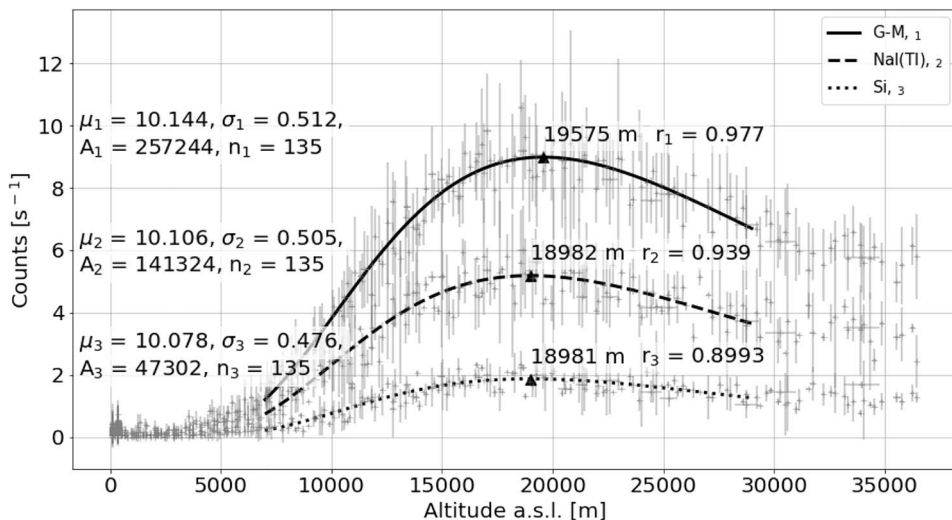


Figure 6. Log-norm fit of ionising radiation measured data with calculated R-P maxima for different detectors (G-M tube, NaI(Tl) scintillator and silicon PIN diode). Measured data are depicted with grey colour as mean from 5 minutes of measurement with 2σ standard errors.

Table 2. Statistical testing of the hypotheses about the equality of the mean values of the log-normal distributions used in Figure 6.

Hypothesis	Statistical criterion t_{exp}	Statistical significance α	Critical value $t_v(\alpha/2)$	Decision
$H_0 : \mu_1 = \mu_2$	0.614	0.01 0.3	2.594 1.038	Accept Accept
$H_0 : \mu_1 = \mu_3$	1.097	0.01 0.3	2.594 1.038	Accept Reject
$H_0 : \mu_2 = \mu_3$	0.469	0.01 0.3	2.594 1.038	Accept Accept

CONCLUSION

Flight FIK-6 was unique due to radiation measurements of the R-P maximum using three different types of radiation detectors. Altitude of the R-P maximum was about 19 km and slightly differed based on the detector type: G-M counter measured the highest value (19 575 m), scintillation detector measured the lower value (18 982 m), and silicon PIN diode the lowest value (18 981 m).. It was shown that this difference is statistically unimportant, except for one case with high significance level 30%. An increased accuracy in evaluation of the altitude of R-P maximum is expected when the temperature corrections will be applied. A telemetric system TF-ATMON has been developed. It enables data recording, pre-flight instruments control and their monitoring during flight. Thanks to the availability of different

communication interfaces on the basic avionics, the use of various alternative detectors is simplified. The technology at the same time improves the possibilities of a quick location of a balloon gondola after its landing. It is therefore possible to carry out even experiments requiring a very short period till recovering the balloon after the flight. The system was successfully tested during FIK-6 flight.

FUNDING

Open Programme Research Development Education, Ministry of Education, Youth and Sports (MEYS), under the project CRREAT, Reg. No. [CZ.02.1.01/0.0/0.0/15_003/0000481] and Czech Technical University (CTU) internal project SGS, Reg. No. [SGS20/181/OHK3/3T/13].

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